Hugo Marquerie 14/12/2025

El conjunto de ceros de una familia de polinomios es igual al conjunto de ceros del ideal que genera

Lema 1. *Sea K un cuerpo y $\mathcal{F} \subset K[x_1, \ldots, x_n]$ no vacío, entonces $\mathbb{V}(\mathcal{F}) = \mathbb{V}(\langle \mathcal{F} \rangle)$.*

***Demostración*:** Veamos ambos contenidos:

$\boxed{\supset}$ Viene dada por la `obs-con-ceros-ideal-generado-contenido-trivial/`Observación 1.

$\boxed{\subset}$ Si $\mathbb{V}(\mathcal{F}) = \varnothing$, entonces $\mathbb{V}(\langle \mathcal{F} \rangle) = \varnothing$ y hemos terminado. Supongamos que $\mathbb{V}(\mathcal{F}) \neq \varnothing$. Sea $a = (a_1, \ldots, a_n) \in \mathbb{V}(\mathcal{F})$, entonces $\forall p \in \mathcal{F} : p(a) = 0$. Sea $q \in \langle \mathcal{F} \rangle$, entonces

$$\exists\, p_1, \ldots, p_r \in \mathcal{F}, s_1, \ldots, s_r \in K[x_1, \ldots, x_n] : q = \sum_{i=1}^{r} s_i p_i \quad (\texttt{lem-ideal-generado/}\text{Lema 1}).$$

Por tanto, $q(a) = \sum_{i=1}^{r} s_i(a)\, p_i(a) = \sum_{i=1}^{r} s_i(a) \cdot 0 = 0$, luego $a \in \mathbb{V}(\langle \mathcal{F} \rangle)$. ■

Referenciado en

- `prop-variedad-algebraica-afin-ideal`

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